Yiğit Sarp Avcı

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Education

Boğaziçi University – B.Sc. in Computer Engineering
09/2023 – Expected 06/2028

- GPA: 2.93/4.00
- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing).
- Volunteering: Volunteering Division Member at Industrial Engineering Club (BÜYAK); educational mentor to high school students.

Kabataş Boys' High School
09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Clubs.
- Volunteering: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

Experience

Backend Developer Intern at DefineX
09/2026 – Present

- Contributing to scalable enterprise software solutions, robust backend architecture, RESTful API design, and system optimizations in a dynamic enterprise environment.

Site Reliability Engineering Intern at Akbank
07/2026 – 09/2026

- Contributed to the reliability, scalability, and performance of mission-critical banking systems within the SRE team. Engaged in Linux systems engineering, operational monitoring, and high-availability systems management.

Software Engineering Intern at Smash Mate
06/2026 – 07/2026

- Contributed to a microservices-oriented backend (Java, Quarkus) and mobile frontend application (React Native) during the pre-launch phase. Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments. Collaborated in agile workflows, code reviews, and system optimizations, directly supporting the product's official launch.

Technical Projects

- **Algorithmic Trading Simulator** — Real-time simulator executing Z-Score mean reversion with volatility sizing.
- **Financial Analytics Dashboard** — Platform featuring a continuous limit-order matching engine & risk models.
- **Restaurant Floor Plan SaaS** — Multi-tenant platform with interactive floor plan builder, table merging & queues.
- **Enterprise Admin Dashboard** — Internal console with role-based access control, server metrics & user workflows.
- **Full-Stack FAQ Module** — Knowledge base with SWR optimistic UI updates, database indexing, RBAC & API tests.
- **Linux IPC Game Subsystem** — 60 FPS architecture with non-blocking FIFOs, POSIX signals, dlopen plugins & buffers.
- **x86-64 CPU Scheduler** — Zero-libc simulator in assembly using raw Linux syscalls (FCFS, SJF, SRTF, PF, RR).
- **Bare-Metal RISC-V Simulator** — RV32I flight pathfinding on Spike with custom linker memory mapping & greedy routing.
- **CoreMark Performance Profiling** — Architectural study evaluating throughput across compiler flags, ILP & ISAs.
- **Unix Command Interpreter** — C-based shell featuring a custom backtracking recursive descent parser & modular state.
- **Functional Language Interpreter** — Custom engine supporting first-class closures, currying, AST evaluation & dual scoping.
- **Graph Network Analyzer** — Network analysis tool identifying articulation points & Tarjan's SCCs in $O(V + E)$ time.
- **Marketplace Matching Simulation** — Platform leveraging custom max-heaps for $O(\log N)$ priority candidate allocation.
- **React Native Minesweeper** — Mobile game featuring iterative BFS flood-fill, safe first-click, chording & haptics.
- **2048 WebAssembly Game** — C++/Qt6 implementation featuring directional grid normalization & infinite undo history.
- **State-Machine Card Game** — Terminal survival game driven by a deterministic state machine & event queues.

Skills, Tools & Languages

- **Programming Languages:** Python, Java, C, C++, TypeScript, JavaScript, SQL, x86-64 Assembly, RISC-V Assembly
- **Technologies & Frameworks:** Next.js, React Native, Quarkus, FastAPI, Express, Tailwind CSS, Prisma, PostgreSQL, Redis, Docker, WebSockets, WebAssembly, Qt6
- **Tools & Software:** Git/GitHub, Linux/POSIX, AI-Assisted Development, MS Office
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)
- **Hobbies:** Wakeboarding, Kiteboarding, Surfing, Bouldering, Running, Fitness, Traveling